AIRBNB PRICING, LOCATION AND LISTING CHARACTERISTICS INVESTIGATION

TASK 4 AND TASK 5: LIMITATIONS AND FINAL ANALYTICAL REPORT

PART A: TASK 4 – INVESTIGATE LIMITATIONS

1. Introduction

The purpose of this section is to identify and discuss limitations that could affect the accuracy, reliability and interpretation of the Airbnb market analysis.

Although the supplied datasets provide useful information about advertised accommodation prices, property characteristics and listing-level review scores, several data-quality and methodological limitations must be considered.

These limitations include missing listing attributes, currency differences, unequal listing counts across cities, duplicate review identifiers, different levels of review activity, missing district information, potentially influential extreme prices and the absence of actual booking and occupancy records.

Understanding these limitations is important because they influence how confidently the research findings can be interpreted and generalised.

2. Missing listing attributes

Some records contained missing information in important listing attributes.

Examples identified during the investigation included:

- Four listings with missing city information.
- Four listings with missing bedroom information.
- Missing review scores for some listings.
- Incomplete district information.

Missing values reduce the amount of information available for analysis. For example, listings without bedroom information cannot be included in an analysis examining the relationship between bedroom numbers and advertised prices.

Similarly, listings without review scores cannot be included in analyses comparing average ratings between accommodation categories.

Missing information may also introduce bias if certain types of listings or cities are more likely to have incomplete records.

Effect on findings: Results calculated using only complete records may not fully represent all listings in the original dataset.

Recommendation: Future data collection should improve the completeness of important listing attributes. Missing values should also be investigated before deciding whether to exclude, retain or impute them.

3. Currency differences

The Listings dataset contains accommodation information from ten international cities, including Cape Town, Sydney, Mexico City, Istanbul, Rome, Rio de Janeiro, Bangkok, Paris, New York and Hong Kong.

These cities may use different local currencies.

Directly comparing raw advertised prices across these cities could produce misleading conclusions. For example, a numerical price of 1,000 in one currency does not necessarily represent the same accommodation cost as 1,000 in another currency.

Exchange rates also fluctuate over time, which means international comparisons require a consistent currency conversion method.

Effect on findings: Raw international price comparisons could incorrectly suggest that accommodation in one city is more expensive than accommodation in another.

Recommendation: Future international price comparisons should convert prices into a common reference currency using appropriate exchange rates and a consistent reference date.

For the main statistical price hypotheses, Cape Town listings were selected to avoid comparing prices across different currencies.

4. Unequal listing counts across cities

The number of listings varied considerably between the ten cities.

City	Number of listings
Cape Town	513
Sydney	484
Mexico City	430
Istanbul	323
Rome	310
Rio de Janeiro	276
Bangkok	235
Paris	196
New York	122
Hong Kong	60

Cape Town had the highest number of listings, while Hong Kong had the lowest.

Unequal sample sizes can affect the reliability and representativeness of comparisons. Cities with larger samples contribute more observations, while smaller samples may be more sensitive to individual unusual listings.

Effect on findings: Results from cities with fewer listings may be less stable and may not represent the wider accommodation market.

Recommendation: Future studies should aim to obtain more balanced and representative samples across cities.

5. Duplicate review identifiers

The `review_id` field is intended to uniquely identify individual reviews.

If duplicate identifiers exist, they could indicate duplicated records or problems during data extraction and processing.

Duplicate records may cause review activity to be overstated because the same review could be counted more than once.

This could distort calculations of review counts per listing and affect conclusions about listing activity.

The investigation also identified a separate and significant relationship problem between the Listings and Reviews datasets:

- Listings records: 2,953.
- Reviews records: 10,000.
- Unique listing identifiers in Reviews: 7,084.

- Matching review records: 62.
- Unmatched review records: 9,938.

This suggests that the two files may have originated from different extracts, versions or sources.

Effect on findings: The mismatch prevents reliable linking of review activity to listing characteristics and city information.

Recommendation: Review identifiers should be checked for uniqueness, and the listing identifiers in both datasets should be reconciled before conducting further joined analysis.

Importantly, the available results do not establish that duplicate review identifiers were actually found. This remains a data-quality check requiring confirmation.

6. Different levels of review activity

The Reviews dataset contained 10,000 records with dates ranging from 3 June 2018 to 2 June 2019.

The monthly distribution was:

Month	Number of reviews
June 2018	1,336
September 2018	1,566
November 2018	2,009
January 2019	1,591
April 2019	1,604
June 2019	1,894

June had a combined total of 3,230 reviews across the two years.

The review dates were unevenly distributed, with records concentrated on Sundays, Mondays and Tuesdays.

Different levels of review activity may be influenced by listing age, guest reviewing behaviour, data completeness and other factors.

A listing with more reviews may appear more active than a listing with fewer reviews, but review counts do not necessarily indicate how many bookings or occupied nights a property received.

Effect on findings: Review counts may not provide a representative picture of activity across all listings or throughout the full period.

Recommendation: Future research should use complete and consistent review records over a longer period and should distinguish review activity from actual booking and occupancy measures.

7. Missing district information

Some listings did not contain district information.

District information is useful for examining geographical differences in advertised prices and accommodation characteristics within cities.

For example, accommodation in a central business district may have different advertised prices from accommodation in a residential or suburban area.

Missing district information limits the ability to conduct detailed neighbourhood-level comparisons.

Effect on findings: The research may overlook important local pricing patterns and geographical differences.

Recommendation: Future datasets should include complete and standardised district information.

8. Potentially influential extreme prices

Cape Town's advertised price distribution showed a wide range of prices.

Measure	Value
Minimum price	152
Median price	625
Mean price	738.73
Maximum price	3,920
Standard deviation	494.45

The mean was higher than the median, suggesting that relatively expensive listings may be influencing the average.

Extreme values can influence statistical summaries and relationships. For example, a small number of expensive listings could increase the average advertised price and create a misleading impression of the typical accommodation price.

However, unusually high prices may represent genuine premium accommodation rather than incorrect data.

Effect on findings: Extreme prices may affect average-price comparisons and statistical relationships.

Recommendation: Extreme values should be investigated before being removed. Median prices and other robust measures should be reported alongside means.

9. Absence of actual booking and occupancy records

The supplied datasets contain advertised listing information and review records. They do not contain actual booking transactions, occupancy rates, cancellations or accommodation revenue.

Advertised prices indicate what hosts are asking guests to pay, not necessarily the amounts guests actually pay.

Similarly, a listing with many reviews is not necessarily fully occupied, and a listing with a high advertised price is not necessarily generating high revenue.

Without booking and occupancy information, the research cannot determine:

- Actual accommodation demand.
- Occupancy rates across cities.
- Revenue generated by individual listings.
- Whether higher advertised prices lead to fewer bookings.
- Which room types generate higher financial returns.
- The relationship between actual booking activity and advertised prices.

Effect on findings: The research can describe advertised market conditions but cannot establish actual market performance or profitability.

Recommendation: Future research should incorporate booking records, occupancy information, cancellations and revenue data.

10. Distinguishing association from causation

An important limitation of the research is that statistical relationships do not automatically demonstrate cause and effect.

For example, Hypothesis 2 identified a positive association between accommodation capacity and advertised price in Cape Town.

The Spearman correlation coefficient was 0.3013, indicating that listings accommodating more guests generally tend to have higher advertised prices.

However, this does not prove that increasing accommodation capacity directly causes the advertised price to increase.

Other factors, such as property size, location, amenities and property quality, may also influence advertised prices.

Similarly, the observed difference in ratings between entire-place and private-room listings does not prove that room type itself causes differences in guest satisfaction.

Effect on findings: The research identifies associations and differences but cannot establish causal relationships.

Recommendation: Further research should include additional variables and, where appropriate, longitudinal or causal research methods.

11. Overall conclusion for Task 4

The Airbnb datasets provide useful information for examining advertised prices, listing characteristics and review ratings across the selected cities. However, several limitations affect the reliability and generalisability of the findings.

Missing listing attributes, unequal city sample sizes, currency differences, extreme prices and incomplete geographical information restrict the comparisons that can be made.

The Reviews dataset also presents challenges because of the potential for duplicate identifiers, uneven review activity and the substantial mismatch between listing identifiers in the Listings and Reviews datasets.

Furthermore, the absence of actual booking and occupancy information means that the research cannot measure actual accommodation demand, revenue or profitability.

Despite these limitations, the research provides useful descriptive insights into the supplied Airbnb listings. The findings should be interpreted cautiously, with particular attention given to data quality, sample size and the distinction between association and causation.

PART B: TASK 5 – PRODUCE AN ANALYTICAL REPORT

AIRBNB PRICING, LOCATION AND LISTING CHARACTERISTICS INVESTIGATION

Executive Summary

This report investigates Airbnb advertised prices, listing characteristics, room types and review ratings across ten international cities.

The purpose is to understand how accommodation characteristics relate to advertised prices and what patterns can be identified from the supplied Airbnb data.

The investigation used two main datasets: Listings.csv and Reviews.csv. The Listings dataset contained 2,953 records and 33 columns, while the Reviews dataset contained 10,000 records and four columns.

Exploratory data analysis identified differences in listing numbers across cities, variations in room types and property categories, missing listing information and differences in review scores.

Cape Town was selected for the main price-related hypothesis tests to avoid comparing raw advertised prices across different currencies.

Three statistical tests were conducted using Cape Town listings. The results showed statistically significant differences in advertised prices across room types, a positive association between accommodation capacity and advertised price, and a statistically significant but small difference in review-rating distributions between entire-place and private-room listings.

A significant data-quality limitation was also identified in the Reviews dataset: only 62 of its 10,000 records matched listing identifiers in the Listings dataset.

Consequently, review activity could not reliably be linked to listing characteristics or city information.

The report concludes that listing characteristics are associated with differences in advertised prices and review ratings. However, the findings should be interpreted cautiously because the datasets do not contain actual booking, occupancy or revenue information.

---

1. Research Problem and Objectives

1.1 Research problem

The primary research problem was:

How do advertised Airbnb listing prices vary across different cities, room types and accommodation characteristics?

Airbnb listings differ in location, accommodation capacity, property type and room type. These characteristics may be associated with differences in advertised prices and guest review ratings.

Understanding these relationships can help identify patterns in the Airbnb accommodation marketplace and provide useful information for hosts, accommodation providers and market researchers.

However, differences in local currencies, unequal listing counts and incomplete information may affect the reliability of international comparisons.

1.2 Research objectives

The objectives of this investigation were to:

1. Examine the distribution of Airbnb listings across the selected international cities.
2. Investigate advertised prices across cities and room types, while accounting for currency differences.
3. Examine property categories and accommodation capacity.
4. Investigate whether accommodation capacity is associated with advertised prices.
5. Examine listing-level review scores and differences between selected property groups.
6. Investigate review activity over time.
7. Identify missing values, unusual observations and data-quality issues.
8. Conduct statistical hypothesis tests to investigate selected relationships.
9. Provide recommendations based on the findings and limitations.

1.3 Research scope

The supplied Listings dataset covered ten cities:

- Cape Town
- Sydney
- Mexico City
- Istanbul
- Rome
- Rio de Janeiro
- Bangkok
- Paris
- New York
- Hong Kong

Cape Town was selected for the three main statistical hypotheses because it had the largest number of listings in the supplied dataset and allowed price comparisons within one local currency.

---

2. Dataset Inspection and Preparation

2.1 Dataset overview

Four supplied files were considered during the initial investigation: two main datasets and their accompanying data dictionaries.

The two main datasets were:

Dataset	Number of records	Number of columns
Listings	2,953	33
Reviews	10,000	4

The Listings dataset contained information about accommodation properties, hosts, locations, room types, advertised prices, capacity and review scores.

The Reviews dataset contained individual review identifiers, listing identifiers, reviewer identifiers and review dates.

2.2 Important variables

The main Listings variables included:

Variable	Description
listing_id	Unique listing identifier
host_id	Host identifier
city	City where the listing is located
district	District or local area
property_type	Type of property
room_type	Type of accommodation offered
accommodates	Number of guests the listing accommodates
bedrooms	Number of bedrooms
price	Advertised listing price
amenities	Available accommodation amenities
review_scores_rating	Overall listing review rating
host_response_time	Host response-time category
host_response_rate	Host response rate
instant_bookable	Whether instant booking is available

The Reviews dataset included:

- listing_id
- review_id
- date
- reviewer_id

2.3 Data preparation

The following preparation activities were undertaken:

- Inspected dataset dimensions and column names.
- Examined data types and numerical and categorical variables.
- Investigated listing and review identifiers.
- Examined missing values.
- Checked advertised price values for unusual or invalid observations.
- Converted review dates into datetime format.
- Created additional date components for analysing review activity.
- Converted selected numerical variables into appropriate numeric data types.
- Created separate working copies for analysis to preserve the original datasets.

The bedrooms column initially contained object/string values rather than a fully numeric data type. It was converted into a numeric working column for analysis. Four missing bedroom values were identified.

Four listings also had missing city information.

2.4 Review-to-listing relationship

A significant issue was identified when validating the relationship between the Listings and Reviews datasets.

The investigation found:

- Listings records: 2,953.
- Reviews records: 10,000.
- Unique listing identifiers in Reviews: 7,084.
- Matching review records linked to Listings: 62.
- Unmatched review records: 9,938.

This indicates that the two datasets may have originated from different extracts, versions or data sources.

The relationship could therefore not be treated as reliable. Review activity was not joined to listing characteristics for city-level or property-level conclusions.

This issue was retained as an important data-quality limitation rather than attempting to force an unreliable relationship.

---

3. Exploratory Data Analysis

3.1 Listing distribution across cities

The distribution of listings was unequal across the ten cities.

City	Listings
Cape Town	513
Sydney	484
Mexico City	430
Istanbul	323
Rome	310
Rio de Janeiro	276
Bangkok	235
Paris	196
New York	122
Hong Kong	60

Cape Town had the highest number of supplied listings, followed by Sydney and Mexico City. Hong Kong had the lowest number.

This indicates that the dataset does not contain an equal representation of all cities. Consequently, results from cities with smaller samples should be interpreted cautiously.

3.2 Advertised price distribution

Advertised prices varied across listings.

For Cape Town, the price summary was:

Measure	Value
Number of listings	513

Mean price	738.73
Median price	625
Minimum price	152
Maximum price	3,920
Standard deviation	494.45

The mean was higher than the median, suggesting that relatively expensive listings may be influencing the average price.

The median provides a useful additional measure because it is less sensitive to unusually high prices.

International price comparisons require currency standardisation. Therefore, raw prices from different countries were not treated as directly comparable.

3.3 Room types and property categories

Cape Town contained the following room types:

Room type	Listings
Entire place	250
Private room	249
Hotel room	8
Shared room	6

Entire-place and private-room listings accounted for most of the Cape Town sample.

The small numbers of hotel-room and shared-room listings mean that conclusions involving these categories should be treated cautiously.

Property categories were also examined. For example, private rooms in houses represented a substantial category in the supplied property data.

3.4 Missing values and unusual observations

Missing values were identified in several listing attributes, including:

- Four missing city values.
- Four missing bedroom values.
- Missing review scores for some listings.
- Incomplete district information.

The Cape Town price analysis found no missing prices and no zero or negative prices among the 513 listings used for the first hypothesis.

Potentially influential high prices were investigated through descriptive statistics. Extreme values were not automatically removed because they may represent genuine premium accommodation.

3.5 Listing-level review scores

The average review scores across the Listings dataset were:

Review measure	Average
Overall rating	94.61
Accuracy	9.68
Cleanliness	9.49
Check-in	9.82
Communication	9.78
Location	9.64
Value	9.53

The overall rating was measured on a 0–100 scale, while the other review dimensions used a 0–10 scale. Therefore, the values should not be directly compared without accounting for their different scales.

Check-in had the highest average score among the 0–10 review dimensions, while value and cleanliness had comparatively lower averages.

These results describe listing-level review scores and do not represent individual review-level ratings.

3.6 Review activity over time

The Reviews dataset contained 10,000 records, with dates ranging from 3 June 2018 to 2 June 2019.

The monthly review counts were:

Month	Reviews
June 2018	1,336
September 2018	1,566
November 2018	2,009
January 2019	1,591

April 2019	1,604
June 2019	1,894

June had 3,230 reviews when the two June periods were combined.

The data also showed an unusual concentration of recorded review dates on Sundays, Mondays and Tuesdays.

These patterns describe the supplied review records only. They should not be interpreted as evidence of booking demand or general guest behaviour because the review dataset may be incomplete and does not cover every calendar month.

3.7 Relationships between selected variables

Spearman correlation analysis was used to examine selected relationships in the Listings dataset.

The overall dataset produced the following results:

Variable relationship	Spearman correlation
Price and accommodates	0.0892
Price and overall rating	0.1279
Accommodates and overall rating	0.0153

These results suggest weak positive associations between price and accommodation capacity, and between price and overall rating. The relationship between accommodation capacity and overall rating was close to zero.

The bedroom variable had very limited variation in the supplied data, with 2,949 valid records showing a value of 1 and four missing values. Therefore, a meaningful correlation analysis involving bedrooms could not be performed.

---

4. Research Questions and Hypotheses

Three hypotheses were formulated using the Cape Town Listings data.

4.1 Hypothesis 1: Advertised prices differ by room type

Research question: Do advertised prices differ across room types in Cape Town?

- Null hypothesis (H_0): There is no statistically significant difference in advertised price distributions across room types.
- Alternative hypothesis (H_1): At least one room type has a statistically significant difference in advertised price distribution.

Variables:

- Independent variable: room_type
- Dependent variable: price

4.2 Hypothesis 2: Accommodation capacity is associated with advertised price

Research question: Is the number of guests a listing accommodates associated with its advertised price?

- Null hypothesis (H_0): There is no significant association between accommodation capacity and advertised price.
- Alternative hypothesis (H_1): There is a significant association between accommodation capacity and advertised price.

Variables:

- Independent variable: accommodates
- Dependent variable: price

4.3 Hypothesis 3: Review ratings differ by room type

Research question: Do overall review ratings differ between entire-place and private-room listings in Cape Town?

- Null hypothesis (H_0): There is no statistically significant difference in overall review-rating distributions between the two room types.
- Alternative hypothesis (H_1): There is a statistically significant difference in overall review-rating distributions between the two room types.

Variables:

- Independent variable: room_type
- Dependent variable: review_scores_rating

---

5. Statistical Methods and Assumptions

A significance level of 5% (α = 0.05) was used for all three hypothesis tests.

5.1 Kruskal-Wallis test

The Kruskal-Wallis test was used for Hypothesis 1 because it compares the distributions of a numerical variable across three or more independent groups without requiring normally distributed observations.

Its main assumptions include:

- Independent observations.
- Independent comparison groups.
- A numerical or ordinal dependent variable.
- Reasonably similar distribution shapes if interpreting the result specifically as a difference in medians.

5.2 Spearman rank correlation

Spearman correlation was used for Hypothesis 2 because it measures the strength and direction of a monotonic relationship between two variables.

Its main assumptions include:

- Both variables are numeric or appropriately ranked.
- Observations are independent.
- The relationship being investigated is reasonably monotonic.
- The data can be meaningfully ranked.

5.3 Mann-Whitney U test

The Mann-Whitney U test was used for Hypothesis 3 because it compares two independent groups without requiring normally distributed observations.

Its main assumptions include:

- Independent observations.
- Two independent groups.
- An ordinal or numerical dependent variable.
- Similar distribution shapes if the result is interpreted specifically as a difference in medians.

The tests identify statistical associations or differences, not causal relationships.

---

6. Results and Interpretations

6.1 Hypothesis 1: Price differences across room types

Test: Kruskal-Wallis

Measure	Result
Test statistic	48.0522
P-value	0.0000000002076
Effect size (ε^2)	0.0885

Since the p-value was below 0.05, the null hypothesis was rejected.

The results indicate a statistically significant difference in advertised price distributions across room types in Cape Town.

The descriptive statistics showed:

Room type	Listings	Mean price	Median price
Entire place	250	816.61	750
Hotel room	8	893.13	875
Private room	249	661.87	519
Shared room	6	477.50	240

The effect size of approximately 0.089 indicates a modest-to-moderate difference in the distributions.

Interpretation: Room type is associated with differences in advertised prices. However, the test does not identify exactly which pairs of room types differ significantly. A post-hoc test would be needed for that purpose.

The small hotel-room and shared-room samples also limit the reliability of comparisons involving these categories.

Hypothesis decision: Reject H_0.

6.2 Hypothesis 2: Accommodation capacity and advertised price

Test: Spearman rank correlation

Measure	Result
Correlation coefficient (ρ)	0.3013
P-value	0.000000000003147

Sample size	513

Since the p-value was below 0.05, the null hypothesis was rejected.

The positive correlation coefficient indicates a weak-to-moderate positive association between accommodation capacity and advertised price.

Interpretation: Listings that accommodate more guests generally tend to have higher advertised prices. However, accommodation capacity alone does not explain all price differences.

Other factors, such as property size, location, amenities and property type, may also influence advertised prices.

Hypothesis decision: Reject H_0.

6.3 Hypothesis 3: Review ratings by room type

Test: Mann-Whitney U

Measure	Result
U statistic	22,318.5
P-value	0.03228
Rank-biserial effect size magnitude	0.1221
Entire-place sample	221
Private-room sample	180

Since the p-value was below 0.05, the null hypothesis was rejected.

The descriptive results were:

Measure	Entire place	Private room
Mean rating	96.40	94.52
Median rating	98	97
Rated listings	221	180

Entire-place listings had slightly higher average and median ratings.

The effect size was small, indicating that although the difference was statistically significant, its practical magnitude was limited.

Interpretation: The two room-type groups have statistically different rating distributions, but the observed difference is relatively small. The result does not establish that room type causes differences in guest satisfaction.

Hypothesis decision: Reject H_0.

6.4 Summary of hypothesis results

Hypothesis	Statistical method	P-value	Decision
Price differs by room type	Kruskal-Wallis	2.076×10^{-10}	Reject H_0
Capacity associated with price	Spearman correlation	3.147×10^{-12}	Reject H_0
Ratings differ by room type	Mann-Whitney U	0.03228	Reject H_0

All three hypotheses produced statistically significant results at the 5% significance level.

However, the practical importance varied. The accommodation-capacity relationship was weak to moderate, while the rating difference had a small effect size.

---

7. Visualisations

Visualisations were used to support the exploratory analysis and make the results easier to interpret.

The following visualisations were created:

7.1 Median advertised price by accommodation capacity

A bar chart compared median advertised prices across different accommodation capacities.

Purpose: To show how typical advertised prices vary according to the number of guests a listing accommodates.

7.2 Median advertised price by room type

A bar chart compared median advertised prices across room types.

Purpose: To highlight price differences between entire-place, private-room and other accommodation categories.

7.3 Average review rating by room type

A bar chart compared average listing-level review ratings across room types.

Purpose: To examine whether different accommodation categories display different average review scores.

7.4 Spearman correlation heatmap

A correlation heatmap displayed the relationships between:

- Advertised price.
- Accommodation capacity.
- Overall review rating.

Purpose: To make the direction and strength of the selected relationships easier to understand.

7.5 Review activity over time

Monthly and date-based summaries were used to examine the distribution of review records over time.

Purpose: To identify periods with higher and lower numbers of recorded reviews.

The visualisations support descriptive interpretation, but they do not independently establish statistical significance or causation.

Figure placement: Insert the actual exported charts from the Python analysis in this section. Give each chart a figure number, descriptive title and short interpretation.

---

8. Limitations

The investigation identified several limitations that affect the interpretation of the findings.

8.1 Missing listing attributes

Missing city, district, bedroom and review-score information reduced the number of complete observations available for certain analyses.

If missing information is concentrated in particular listing categories, results based on complete records may be less representative.

8.2 Currency differences

The dataset covers international cities with different local currencies.

Raw advertised prices cannot be directly compared across these cities without currency standardisation using appropriate exchange rates and a consistent reference date.

The main price-related hypothesis tests were therefore restricted to Cape Town.

8.3 Unequal listing counts across cities

Cape Town contained 513 listings, while Hong Kong contained only 60.

Unequal sample sizes may affect the stability and representativeness of city-level comparisons.

8.4 Duplicate review identifiers

Review identifiers should be unique. Duplicate identifiers, if present, could indicate duplicated records and cause review activity to be overstated.

Identifier uniqueness should therefore be confirmed before relying on review counts. The supplied results do not establish that duplicates were found.

8.5 Different levels of review activity

Review dates were unevenly distributed across the supplied period.

The number of reviews may also depend on listing age, guest reviewing behaviour and dataset completeness.

Review counts should not be treated as direct measures of bookings, demand, occupancy or revenue.

8.6 Missing district information

Incomplete district information limits detailed geographical comparisons within cities.

This makes it more difficult to identify neighbourhood-level pricing patterns.

8.7 Potentially influential extreme prices

High advertised prices may influence mean calculations and statistical relationships.

Although extreme values can affect results, they may represent genuine premium properties. They should be investigated rather than automatically removed.

8.8 Absence of actual booking and occupancy records

The datasets contain advertised listing information and review records, but not actual booking transactions, occupancy rates, cancellations or revenue.

Therefore, the research cannot establish actual demand, financial performance or profitability.

8.9 Association versus causation

The statistical tests identify relationships and differences, but they do not establish cause and effect.

For example, the positive association between accommodation capacity and price does not prove that increasing the number of guests a property accommodates will cause its price to increase.

Other factors, including location, property size and amenities, may influence the relationship.

The research is observational, and the findings should therefore be interpreted as associations rather than causal effects.

---

9. Recommendations

Based on the analysis and its limitations, the following recommendations are proposed.

9.1 Consider room type when setting advertised prices

The analysis identified statistically significant price differences across room types in Cape Town.

Hosts should consider the type of accommodation they offer when reviewing their advertised prices. However, they should also consider property quality, amenities, location and competing listings.

9.2 Consider accommodation capacity in pricing decisions

The positive association between accommodation capacity and advertised price suggests that capacity is a useful factor to consider when evaluating listing prices.

Hosts should assess their accommodation capacity alongside other property characteristics rather than assuming that capacity alone determines an appropriate price.

9.3 Monitor guest review ratings

Entire-place listings had slightly higher average and median ratings than private-room listings in the Cape Town sample.

Hosts should monitor guest feedback and pay attention to areas such as cleanliness, check-in, communication and value.

These measures may help identify opportunities to improve the guest experience.

9.4 Improve data quality and completeness

Future data collection should aim to ensure that important listing attributes, such as city, district, bedrooms and review scores, are consistently recorded.

Unique identifiers should also be validated to reduce the risk of duplicated or incorrectly linked records.

9.5 Ensure consistency between Listings and Reviews datasets

The substantial mismatch between listing identifiers in the two datasets should be investigated.

Future analysis should use compatible data extracts with verified listing identifiers so that review activity can be accurately linked to listing characteristics.

9.6 Standardise currencies for international comparisons

Future international price comparisons should convert advertised prices into a common reference currency using appropriate exchange rates and a consistent conversion date.

This would make comparisons across cities more meaningful.

9.7 Obtain booking and occupancy information

Future research should include actual booking records, occupancy rates, cancellations and revenue information.

This would allow researchers to investigate real accommodation demand and financial performance rather than relying only on advertised prices and review counts.

9.8 Use more representative samples

Future studies should aim for more balanced and sufficiently large samples across cities and accommodation categories.

This would improve the reliability of comparisons and reduce the influence of very small groups.

---

10. Overall Conclusion

The Airbnb Pricing, Location and Listing Characteristics Investigation identified several patterns in advertised accommodation prices, property characteristics and listing-level review ratings across ten international cities.

The exploratory analysis demonstrated that listing counts varied considerably between cities and that advertised prices and accommodation categories differed across the supplied data.

The statistical hypothesis tests provided evidence of three relationships within Cape Town. Advertised price distributions differed across room types, accommodation capacity was positively associated with advertised price, and overall review-rating distributions differed between entire-place and private-room listings.

Although all three results were statistically significant, their practical importance varied. The analysis also highlighted several data-quality limitations, particularly missing listing information, unequal city samples and the substantial mismatch between listing identifiers in the Listings and Reviews datasets.

The absence of actual booking and occupancy records further limits the conclusions that can be drawn about accommodation demand and financial performance.

Overall, the research provides useful insights into the characteristics associated with advertised Airbnb prices and listing-level review ratings. However, the findings should be interpreted within the limitations of the supplied datasets. More complete, consistent and representative data would be required to develop a deeper understanding of Airbnb market performance.
